

b	1. The force causes <u>repulsion</u> between the ions. (This is because the gradient of the part of graph of that corresponds to $E = \frac{B}{r^8}$ is negative, hence, the force $F = -\frac{dE_P}{dr}$ is positive) 2. The force <u>is significant when the ions are very close to each other, and decreases rapidly as the distance increases.</u> (As seen from the very steep gradient when the ions are close to each other).	B1 B1
ci	1. Horizontal line at -6 eV for total energy, labelled T. 2. Reasonable curve starting and ending at zero, with a maximum point Curve's maximum point is at $(2.75, 2.0)$, since potential energy is minimum at that point. By right, K line should be drawn such that at every point, $KE + PE = 6\text{eV}$.	A1 A1 A1

cii	Minimum value of 2.35×10^{-10} m. Maximum value of 4.25×10^{-10} m	A1 A1
ciii	F is not directly proportional to displacement r. OR the graph is not symmetrical Hence, not simple harmonic.	B1 A1
d	As the lattice is heated, its <u>total energy increases</u>. The <u>total energy</u> of the ions <u>intersects the graph Fig. 7.3 at higher levels</u>, giving rise to <u>larger changes in potential energy</u> during the oscillations. The ions will vibrate between greater values of separation r.	B1 B1 B1